Statement of Work (SOW)
For
Air Force Civil Engineer Center
Explosive Ordnance Disposal
Response Trailers

Combined Synopsis/Solicitation: FA8051-26-R-0005

05 May 2026

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1. **Introduction.** The Air Force Civil Engineer Center, Explosive Ordnance Disposal Division (AFCEC/CXD) requires enclosed trailers to provide Explosive Ordnance Disposal (EOD) teams at Continental United States (CONUS) installations with a dedicated, mobile response capability. These trailers will be pulled by a new prime mover (tow vehicle), which is being acquired separately and is not part of this requirement. The trailers will support emergency response and Defense Support of Civil Authorities (DSCA) missions at CONUS bases.

2. **Scope.** This Statement of Work (SOW) establishes the mandatory technical and manufacturing specifications for approximately 7’ x 20’ enclosed, all-terrain EOD Response Trailers. The Contractor shall manufacture and deliver trailers that strictly conform to all dimensional, structural, electrical, and accessory requirements detailed within this document. The design must support the safe transport of heavy, specialized EOD payloads over improved roadways (paved highways) and unimproved surfaces (gravel, dirt, sand, and flat grass). Upon arrival at a response site, the trailer must function as a self-contained, climate-controlled workspace. The trailers will be fielded to various Continental United States (CONUS) locations as designated by individual Delivery Orders. See the Attachment 1a EOD Response Trailer Fielding Locations document for a list of anticipated delivery sites.

3. **Trailer Specifications - General.** The following specifications outline the general design, compliance, and structural baseline requirements for the EOD Response Trailers. All delivered trailers must incorporate the following features:

   3.1 General Capability: Must be an on- and off-road (all-terrain) capable trailer designed to transport and secure emergency response equipment while providing a secure, climate-controlled workspace during operations.

   3.2 Department of Transportation (DOT) Compliance: Must comply with all applicable DOT regulations to permit legal use on all highways and motorways. This includes all standard DOT marking lights, reflectors, and an illuminated license plate mount.

   3.3 Lighting & Power: All lights (exterior and interior) shall be Light Emitting Diode (LED) only. They must be powered via a standard 7-pin light connector at the front of the trailer, allowing the prime mover (tow vehicle) to power all DOT lights, electric brakes, and three (3) interior loading lights.

   3.4 Cargo Box Design: The trailer cargo box shall be built with the wheels exterior to the cargo box (i.e., "fendered wheel wells") to provide maximum, unimpeded interior floor space.

   3.5 Exterior Color: Trailer exterior walls shall be black or grey.

   3.6 Personnel Door: The curb-side personnel door must feature one window, tinted at 28% Visible Light Transmission (VLT).

   3.7 Front Nose Profile: The trailer must have a flat front nose to accommodate a mounted generator; it must not be rounded or pointed (see Section 4.2 for further details). The driver and passenger side leading corners may be rounded.

4. **Dimensions, Axles, Wheels, DOT Lights, Rear Camera.** This section defines the physical footprint, weight limits, and primary exterior features required for the trailer platform.

   4.0.1 Gross Vehicle Weight Rating (GVWR): Shall not exceed 7,000 lbs.

4.0.2 Trailer Box Dimensions: The interior "box" shall measure 84” in width by 192” in length (7’ x 16’).

4.0.3 Overall Length: Must be between 19.5’ and 20’ from the front hitch to the rear of the trailer structure (excluding handles, latches, or cameras).

4.0.4 Overall Width: Approximately 8.5’ (102”) from fender-to-fender.

4.0.5 Hitch Type: Bumper Pull with a 2 5/16” ball hitch.

4.0.6 Interior Height: Minimum of 84” from floor to ceiling.

4.0.7 Personnel Door: One oversized RV-style personnel door (32” x 72”) on the curb-side, featuring an electronic keypad lock with a key backup (two keys provided).

4.0.8 Entry Step: A manual, single pull-out step (minimum 25” wide) shall be installed directly under the personnel door.

4.0.9 Rear Camera System: A rear-facing camera shall be mounted above the rear door. All power and video cabling must be routed through the interior walls, protected by sheathing, and terminate at the front of the trailer to connect to the prime mover (tow vehicle) display system.

4.1 Wheels, Axles & Ground Clearance:

4.1.1 Wheels and Tires: Shall be aluminum five-lug wheels with 32” all-terrain tires rated for the maximum loaded GVWR of the trailer or greater. A full-size spare tire and matching wheel must be provided and secured inside the trailer.

4.1.2 Axles and Suspension: The trailer shall be equipped with a tandem axle system providing performance equal or superior to industry-standard internal rubber torsion axles. To be considered equal or superior, the system must meet or exceed the following characteristics:

a. Capacity: Minimum 3,500 lb. Gross Axle Weight Rating (GAWR) per axle.
b. Suspension Type: Internal rubber torsion suspension (or a superior independent suspension technology that eliminates leaf springs and enhances shock absorption).
c. Independent Movement: Must provide true independent wheel suspension to optimize handling and stability on uneven terrain.
d. Brake Compatibility: Must be equipped with standard 5-bolt brake flanges and fully compatible with 10" electric brakes.
e. Geometry: Must be configured with a 10° Up starting angle (or an equivalent geometry that achieves the required 16" ground clearance).

4.1.3 Ground Clearance: The completed trailer shall provide a minimum ground clearance of 16 inches, measured from a level surface to the lowest point of the frame, body, or any attached component (excluding tires, wheels, and axle assemblies).

4.2 Exterior Structure:

4.2.1 Frame: Must be a steel tube structural frame coated with a rubberized/asphalt coating designed for a 10-year service life. Wall posts and floor crossmembers shall be 16” on centers.

4.2.2 Tongue: A triple tube tongue with a 2 5/16” ball hitch is required.

4.2.3 Trailer Jack: A single 3,500 lb. electric trailer jack shall be mounted on the center tube behind the hitch. Its battery shall be mounted on the exterior tongue (passenger side), clear of all connections, turning radius, and the generator cabinet.

4.2.4 Exterior Sheeting: Must be aluminum with a thickness of .030 or greater, with Paint Protection Film (PPF) applied on all sides.

4.2.5 Generator Platform: A 36” x 30” generator platform and cabinet must be secured to the triple tube tongue, within 1” of the front trailer wall but not touching it. The platform must include tie-down points to allow for generator removal. The cabinet must feature a weatherproof pass-through for the electrical cable that prevents cable damage.

4.2.6 Stone Guard: A stone guard must cover the lower 24” of the front trailer wall.

4.2.7 Ramp Door: A Super Duty Ramp Door with a strip-flap hinged top is required. The ramp door must be appropriately sized to seal when closed without impacting DOT lights or swing-arm bars. It shall be secured by two swing cam-bar lock assemblies (one on each side) compatible with padlocks.

4.2.8 Rear Stabilizer Jacks: Two 7,500 lb. stabilizer jacks shall be mounted under the rear corners of the trailer, positioned so they do not impede the 16” ground clearance requirement.

4.2.9 Cable Pass-Throughs: Two deluxe, 5” diameter cable pass-throughs with sealable doors (black) are required, separate from the shoreline connection. One shall be located high on the curb side behind the ladder (approximately 12” from the roof); the second shall be located forward of the personnel door at approximately 40” from the floor.

4.2.10 Explosive Placards: Flip-style placards for Class A explosives (1.1, 1.2, 1.3, 1.4, 1.5) shall be mounted and watertight sealed by the manufacturer on all four sides of the trailer. The front placard shall be mounted centered, above the generator cabinet. The side placards shall be mounted as close to the center of the trailer as possible, positioned above the wheel fenders to ensure a clean, professional appearance, and their mounting hardware must not impact any electrical wires or connections. The rear placard shall be mounted as close to the center of the ramp door as possible without being damaged when the ramp is opened during loading and unloading operations.

4.3 HVAC & Roof Mounted Items:

4.3.1 Air Conditioner/Heater: One low-profile (9” to 11” high) 15K AC unit with a 12K heat strip shall be mounted center on the roof-top, powered by the generator or shore power.

4.3.2 RV Vent: One 14” x 14” RV vent with a built-in fan, on/off switch, and crank-to-open cover shall be mounted center on the roof-top, 24” from the forward edge of the trailer, powered by the generator or shore power.

4.3.3 Awning: One 12’ long, 12v powered awning (white top and bottom) shall be installed on the curb side, positioned so it does not impede the personnel door when deployed and powered by the generator or shore power.

4.3.4 Roof Rack: A low-profile roof-top storage rack is required, providing a minimum of 21 sq. ft. of storage capacity (est. 7 ft. wide x 3 ft. long) and capable of holding 400 lbs. or more. The rack shall be bolted to the trailer's crossbar frame in at least four locations, positioned close to but not touching the roof.

4.3.5 Ladder: A personnel ladder shall be bolted to the curb side of the trailer, directly behind the axles. The ladder shall not protrude farther from the side of the trailer than the fender. Rungs must have a non-slip surface but shall not use grip strut or similar materials that could damage equipment or cause injury.

4.4 Electrical Connections:

4.4.1 Code Compliance: All electrical connections, interior and exterior, shall meet or exceed National Electric Code (NEC) Articles 551 & 552 or applicable codes for DOT lighting.

4.4.2 Interior Electrical Box: The trailer shall feature a 120/240V, 50-amp electrical system. The main interior electrical box shall be mounted in a protective enclosure on the driver's side front wall, positioned 6" to 8" above the floor. The box must have sufficient capacity to support all required circuits, plus a minimum of four (4) open spaces for future breaker expansion. All required breakers shall be installed.

4.4.3. Converter/Charger: The system must include a power converter/charger capable of powering all outlets, lights, and the rooftop air conditioner, as well as charging the electric trailer jack battery. The AC/DC Converter shall be mounted adjacent to the breaker box and wired appropriately. The AC/DC Converter must be large enough to accommodate the power draw of the existing systems plus an additional 240/50 hz (should the four expansion slots be used).

4.4.4 Generator Power Cable: The power cable running from the Interior Electrical Box to the generator shall be covered and physically protected inside the trailer, and wired to a weather sealed power plug port, mounted above the rock guard on the front of the trailer - offset to the drivers side of the generator. The Generator power cable shall be long enough to plug into the mounted generator and plug into the power port safely. An all-weather engraved metal label above the power port shall read – GENERATOR.

4.4.5 Generator: A 5.5KW, 50-amp gas generator capable of providing both 110v and 240v power shall be installed and secured in the generator cabinet. The generator must have remote start capability, and at least one engine start key-fob shall be provided.

4.4.6 Shore Power: A separate, weather-sealed power port shall be mounted on the front of the trailer, adjacent to the generator port. An all-weather, engraved metal label reading "SHORE POWER" shall be mounted directly above this port. The contractor shall provide one (1) 100 ft. shore power cable with the required male and female plugs.

4.4.7 Power Port Weatherproofing: Both the Generator power port (4.4.4) and the Shore Power port (4.4.6) shall feature a self-closing (spring-loaded), sealable cover that protects the port when not in use. Both power ports shall be completely enclosed to ensure that when exterior cables are plugged in, weather elements are prevented from entering the interior of the trailer.

4.4.8 Exterior Outlets: Two 110-120v 60Hz duplex outlets in protective, all-weather housings are required: one on the driver's side and one on the curb side, each located 2" above the axle fender.

4.4.9 Interior Outlets: Eight (8) four-plugin outlet boxes (110-120v) with protective covers are required. They shall be spaced as follows: four on the driver's side, two along the front wall, and two along the curb-side wall (between the personnel door and the back wall). All shall be mounted approximately 37" up from the floor (just above and not impeding use of the E-Track).

4.5 Exterior Lighting:

4.5.1 The trailer shall be equipped with a 360-degree LED flood/scene lighting system to illuminate the exterior workspace. The system shall be powered by the 120V AC electrical system (shore power or generator). The lighting system shall be mounted along the roofline or on the roof, angled down appropriately, and consist of the following:

a. Driver's Side: Two (2) LED light bars (18" to 26" in length) providing a minimum of 40,000 Lumens each, located near the top corners of the trailer.
b. Curb Side: Two (2) LED light bars (18" to 26" in length) providing a minimum of 40,000 Lumens each, located near the top corners of the trailer.
c. Rear: Two (2) 4" x 4" LED light pods providing a minimum of 8,400 Lumens each. These pods shall be vertically adjustable and evenly spaced with the rear backup camera located in the center.
d. Front: Two (2) 4" x 4" LED light pods providing a minimum of 8,400 Lumens each. These pods shall be vertically adjustable and evenly spaced from the trailer centerline.

4.5.2 Control Panel: A single control panel for all exterior scene lighting shall be located on the interior, just aft of the personnel door frame.

4.6 Interior Lighting:

4.6.1 Primary Lighting: Eight (8) surface-mounted LED lights providing a minimum of 900 Lumens each, shall be installed on the ceiling in two rows of four for even illumination throughout the workspace. These lights must be dimmable via a switch located near the personnel door and powered by the generator or shore power.

4.6.2 Loading Lights: Three (3) interior LED loading lights providing a minimum of 900 Lumens shall be mounted on the ceiling centerline, beginning three (3') feet in from the rear ramp. These lights must be powered by the prime mover (tow vehicle) or shore power to allow for visibility when the generator is not running.

4.6.3 Loading Light Controls: The loading lights shall be controlled from a panel located near the personnel door and must also have a second, separate switch located near the rear ramp door for ease of use during loading and unloading operations.

4.7 Interior Walls, Ceiling & Flooring:

4.7.1 Insulation: Polyfoam or similar insulation is required in the ceiling and all walls.

4.7.2 Walls & Ceiling: Walls and ceiling shall be finished with white vinyl. Wall material must also be coated with a dry-erase material. All fasteners shall be flush for a clean appearance.

4.7.3 Flooring: The floor and ramp door shall feature a ¾" plywood subfloor covered with a heavy-duty, non-slip, fully-adhered polyvinyl flooring (75 mil minimum thickness) with a raised diamond tread pattern, using G-Floor 75 Diamond Tread or an equivalent product that meets or exceeds its specifications (e.g., 2,000 PSI static load rating per ASTM F970).

4.7.4 Wall E-Track System: Four (4) rows of E-Track rails are required on the driver's side, front, and curbside walls. These shall be installed at the floor line, at 30” and 60” (measured from the floor), and at the roof line. All wall E-Track must be recessed to provide a flush transition from the E-Track to the wall.

4.7.5 Floor E-Track System: Three (3) rows of E-Track are required, running the length of the interior. The tracks must be recessed for a smooth surface. One row shall be installed exactly on the centerline. The remaining two rows shall be offset from the centerline to provide exactly 30” of clear space between the two outer tracks (Note: The current robotic platform weighs 710 lbs. and is 47.3” long by 27.6” wide).

4.8 Other:

4.8.1 Fire Extinguishers: Two (2) 2A-10BC fire extinguishers with Rubber-Strap Vehicle Mounts shall be mounted on the front exterior wall of the trailer above the rock guard. They shall be offset evenly from the generator cabinet so as not to impede its function or the loading/unloading of a generator. They shall not be mounted along the leading edge of the trailer to prevent impacts from the prime mover (tow vehicle).

4.8.2 Weather Sealing: All screws, rivets, wire holes, etc., will be sealed with automotive industry sealants and materials, utilizing automotive industry best practices to prevent water intrusion and provide maximum longevity.

5. Additional Expectations. This section outlines the mandatory requirements for the First Article Inspection, production timelines, shipping coordination, final delivery and acceptance, and the unified warranty requirement.

5.1 First Article Inspection: The production and delivery of a First Article unit is required. An in-person First Article Inspection (FAI) and formal Government approval of one fully completed trailer shall be conducted prior to the commencement of the full production run for any remaining delivery order quantities.

5.2 Production and Inspection Timeline: All trailers ordered under a specific delivery order shall be fully manufactured and ready for Government inspection no later than 180 calendar days after the date of that delivery order. Upon notification from the contractor that production is complete, AFCEC/CXD personnel will conduct a full inspection of all trailers at the manufacturing facility, using this Statement of Work as the minimum inspection criteria. Any discrepancies identified during this inspection must be fully corrected at the manufacturing facility prior to the Government authorizing shipment.

5.3 Final Delivery and Acceptance: After successful completion of the Government inspection at the manufacturing facility, final delivery of each trailer to its designated destination shall occur within 30 calendar days. Shipment to the final destination is the responsibility of the contractor. Anticipated shipping zip codes are provided in the EOD Response Trailer Fielding Locations attachment; complete delivery addresses will be provided on individual delivery orders.

5.3.1 Shipping Coordination: Prior to executing any shipment, the contractor shall obtain and submit a minimum of two (2) competitive freight/shipping quotes to the Contracting Officer for review and approval. All proposed shipping methods and selected carriers must provide online tracking capabilities that are fully accessible by the Government until final delivery is confirmed.

5.3.2 Acceptance: A US Government representative will conduct a final acceptance inspection upon delivery to identify any damage or defects that may have occurred during shipping. Final Government acceptance will be executed only after a successful delivery inspection at the destination site.

5.4 Minimum Warranty Requirement: The delivered trailer must be accompanied by a single, unified "bumper-to-bumper" warranty. This unified warranty shall, at a minimum, cover the entire trailer structure and all installed subsystems against defects in materials and workmanship for a single, comprehensive duration. Fragmented warranties with differing coverage durations for individual components are not acceptable. The Contractor shall be bound by the specific warranty duration and terms proposed and accepted at contract award. All applicable warranty documentation must be provided at the time of delivery.

<u>Attachments:</u>
1a. EOD Response Trailer Fielding Locations